



BIO411 End-semester Exam (2022) IISER Mohali
Date: 30/11/2022 **Duration:** 90 min
Maximum marks: 30

Name: _____ **Roll No:** _____

Instructions:

- 1) Answer all questions in Section A. All questions in this section carry 1 marks.
- 2) In Section B, Questions no 1 is compulsory. From rest answer any 2 questions. Each question carries 5 marks.

Section A

1. Write a peptide sequence which will match the prosite pattern {AL}-G-X(2)-Q-[MIL]-T
2. Gene Ontology defines function in three broad levels: Biological Process, Molecular function and cellular components.
3. In PDB format file, the TER record indicates end of protein atom coordinates and HETATM record consists coordinates of metals, chemical compounds, peptides, water and DNA/RNA
4. Levinthal Paradox is protein would take infinite amount of time to fold if it explores all possible configuration, however, protein folds within seconds.
5. In protein structural classification (SCOP), two proteins are related at the level of 'Fold' if these structures same major secondary structures in the same arrangement and with same topological connections.
6. A protein XP of length 250 residues takes a spherical solid shape of diameter 150 Å. The longest possible α -helix in this structure has 100 number of residues.
7. The context based function annotation can be performed using STRING webserver/database.
8. A pdb structure deposited in the PDB database has 20 models. Based on this, the most likely experimental method of structure determination is NMR.
9. The Jpred/PSIPRED/GOR method can be used for secondary structure prediction and DSSP/STRIDE method can be used for secondary structure assignment.
10. Deby-waller factor is defined as a factor, which quantifies the uncertainty in the atomic position. These are fitted parameter in structure determination to account for the smearing of electron density, which could be because of thermal vibration or motion of atoms. The amount of smearing is proportional to the magnitude of the B-factor.
11. Inverse protein folding protein is defined as the problem of finding one or more sequences compatible with a given fold.
12. The solvent accessible surface area of a protein is defined as Solvent Accessible Surface Area, whereas Connolly surface (molecular surfaces) includes both Contact and reentrant surface.

13. In protein tertiary structure modeling/prediction context, Template refers to **experimentally determined tertiary structure on which sequence is modeled or used for threading.**
14. The secondary structure propensity of Proline (P) amino acid in helix (H) is **0.25** , with below data.
 $n(P) = 1500$; $n(E) = 4500$; $n(H) = 8000$; $n(\text{Total}) = 15000$; $n(P, E) = 900$; $n(P, H) = 200$
15. The metric **relative solvent accessible surface area** can be used to identify whether a residue is buried or exposed in protein tertiary structure.

Section B

1. Use schematic diagram to show dihedral angles ϕ , ψ , ω for ALA-ALA-ALA dipeptide unit. Describe briefly basis used for defining allowed, partially allowed, and disallowed regions in Ramachandran map. Is Ramachandran map obtained for ALA dipeptide is same/similar for all amino acids? Justify your answer. Define χ angle.

You should have used 3 ALA residues to represent ϕ , ψ , ω and these angles are defined for central alanine residue with following definitions:

$$\Phi = C^{i-1}N^iC^i_aC^i$$

$$\Psi = N^iC^i_aC^iN^{i+1}$$

$$\omega = C^{i-1}_aC^{i-1}N^iC^i_a$$

For allowed, partially allowed, and disallowed regions refer to my lecture notes.

Yes, Ramachandran map for all amino acids except glycine is same with restricted map for Proline. It is essentially because steric hindrance responsible for disallowed regions in R-map is restricted to main chain of polypeptide chain and side chain atom C β . Glycine has very broad allowed/partially allowed region in R-map. Proline has restricted Φ angle at -60° .

2. A scientist determined tertiary structure of an enzyme bound with a ligand. The scientist would like to know whether the bound ligand shows binding specificity. Suggest an *in-silico* approach to study protein-ligand interactions and interactions, which potentially defined specificity. Provide detailed parameters to identify intra-molecular interaction, which determines specificity.

The protein-ligand interactions can be studied by analyzing inter-atomic contact criteria of 4.5Å to determine residues, which are interacting with ligand atoms. Hydrogen bonds confer directionality and specificity to the intramolecular interactions in protein-protein, protein-ligand structures. The h-bond parameters needs to be described.

3. You have determined tertiary structure of a globular protein, which binds to GTP but structure does not have GTP bound to it. Describe steps of any one algorithm, which can find potential binding site of GTP.

Describe either LigSite or Surfnets algorithm

4. Is there a unique optimal solution to structural alignment? Discuss briefly various broad types of structural alignment of proteins and also suggest context of their usage. Discuss two metrics used to measure structural similarity between two structures and their limitations, if any.

No, there is no unique optimal solution for sequence independent structural alignment. But for sequence dependent structural alignment there is a unique answer. Aligning protein structures. Discuss sequence dependent and sequence independent structural alignment. RMSD and TM-score are two metrics to evaluate structural similarity. RMSD is a commonly used metric to evaluate similarity between protein structures. RMSD is a global structural similarity parameter that can be used to study conformational change in protein structures and to assess quality of modelled structures. Since RMSD is a function of number of structurally equivalent residues (N), it cannot be used as a general metric to compare RMSD values calculated for pairs of structures of varying length. For instance, the conformational change of 50 residues protein having RMSD of 1 Å is not the same as RMSD of 1 Å for 200 residues.

5. Why L-amino acids do not adopt left-handed helix conformation? Suggest an approach to design amphipathic helix. Describe various features of various helices observed in protein tertiary structures. Can you identify secondary structure based only on ϕ , ψ angles? Justify your answer.

Steric hindrance between C β and backbone CO prevents formation of left handed helix for L-amino acids. Helical wheel to construct amphipathic helix. Describe various types of helices. Usually, ϕ / ψ angle is not sufficient to define secondary structure as these are defined based on backbone hydrogen bonding pattern.